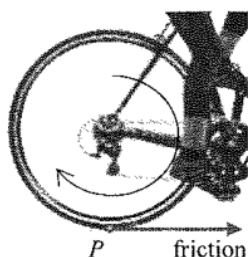


1.	(a)	Heat given out by iron cube = Heat gained by water $0.2 \times 450 \times (T - 33) = 0.6 \times 4200 \times (33 - 25)$ $T = 257\text{ }(^{\circ}\text{C})$	2M	
			1A	
			3	
	(b)	Some water might become steam (immediately). <u>Or</u> Some water in the cup might splash.	1A	
			1A	
			1	
	(c)	It is because some heat is lost to the surroundings through vaporization of water / as latent heat. <u>Or</u> It is because some heat is lost to the surroundings (or polystyrene cup) / The water might not be well stirred, otherwise, the water should have reached a temperature higher than $33\text{ }^{\circ}\text{C}$ (i.e. underestimated).	1A	
			1A	
			1A	
			2	
	(d)	(i) Poor thermal contact between the glass bob of the thermometer and the iron cube.	1A	
			1	
	(ii)	Infra-red (IR) thermometer.	1A	
			1	

2. (a) $m_0 = \text{density} \times \text{volume}$ $= 1.20 \times (1.5 \times 10^{-3})$ $= 1.8 \times 10^{-3} \text{ kg or } 1.8 \text{ g}$	1M 1A 2	
(b) (i) $p_1 V = n_1 R T$ and $p_2 V = n_2 R T$ As $m_0 \rightarrow 4m_0$, $n_2 = 4n_1$ $\therefore p_2 = \frac{n_2 R T}{V} = \frac{4n_1 R T}{V} = 4p_1 = 4.0 \times 10^5 \text{ Pa (or } 4.0 \text{ atm)}$ Or $pV = nRT$ $p \propto n$ (as V, R and T are constant) Since $m_0 \rightarrow 4m_0$, thus $n \rightarrow 4n$, p becomes 4 times the original value ($= 1.0 \text{ atm}$), $p = 4.0 \times 10^5 \text{ Pa (or } 4.0 \text{ atm)}$	1M 1A 1M 1A 2	
(ii) More air molecules hit unit area of surface of wall or bottle / water surface (as $n \rightarrow 4n$ moles of molecules trapped in the same volume), The surface of wall or bottle / water surface is hit more frequently OR rate of change of total momentum, i.e. force, per unit area of wall or bottle / water surface increases, pressure increases.	1A 1A 2	
(iii) $\Delta F = \text{increase in pressure} \times \text{area}$ $= [(4 - 1) \times 10^5] \times 0.014$ $= 4200 \text{ N}$	1M 1A 2	error-carried-forward (e.c.f.) from (b)(i)
(c) (i) The pressure force acting on the water results in its change of momentum (a water jet). Magnitude of the force = rate of change of momentum. According to Newton's third law of motion, an equal but opposite force / reaction / thrust acts on the rocket. (According to Newton's second law of motion,) the reaction force causes the rocket to accelerate / move upwards.	1A 1A 2	
(ii) kinetic energy of (the jet of) water / work done on (the jet of) water	1A 1	

3. (a)



1A

1

- (b) Mechanical power delivered (for work done against the resistive force)

$$P = Fv$$

$$= 17.0 \times 8.0$$

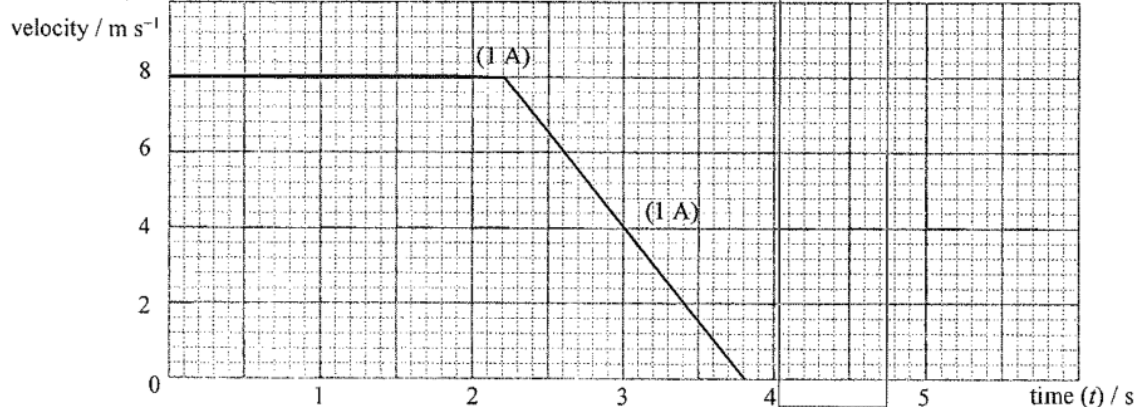
$$= 136 \text{ W}$$

1M

1A

2

- (c) (i)



2

- (ii) Deceleration: $\left| \frac{0-8}{3.8-2.2} \right| = 5.0 \text{ m s}^{-2}$
Resistive force = 65×5.0
 $= 325 \text{ N}$

1M

e.c.f. from (c)(i)

1A

Alternative method:

$$\text{Braking distance} = \frac{1}{2}(8)(3.8 - 2.2) = 6.4 \text{ m}$$

$$Fs = \frac{1}{2}mv^2 - 0$$

$$F(6.4) = \frac{1}{2}(65)(8)^2$$

$$F = 325 \text{ N}$$

1M

1A

2

- (iii) Total distance travelled after $t = 2.0 \text{ s}$
 $= \text{area under the graph from } t = 2.0 \text{ s to } t = 3.8 \text{ s}$
 $= \frac{0.2 + 1.8}{2} \times 8.0 = 8.0 \text{ m}$

1M/1A

$$\text{Distance between the bicycle and the obstacle when the bicycle stops} = 9 - 8$$

$$= 1 \text{ m}$$

1M

1A

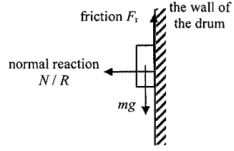
3

- (d) The padded cushion can increase the contact time of the knee and the ground such that a smaller force acts on the knee.

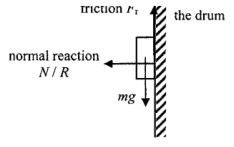
1A

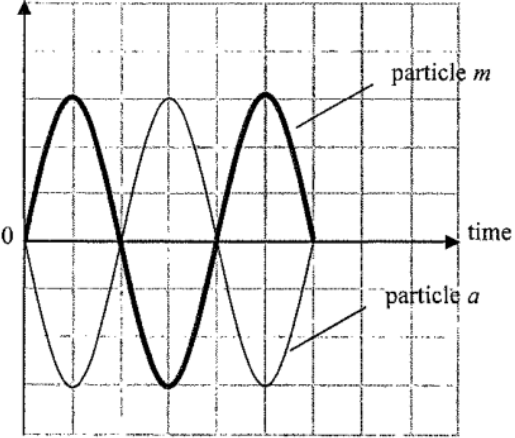
1A

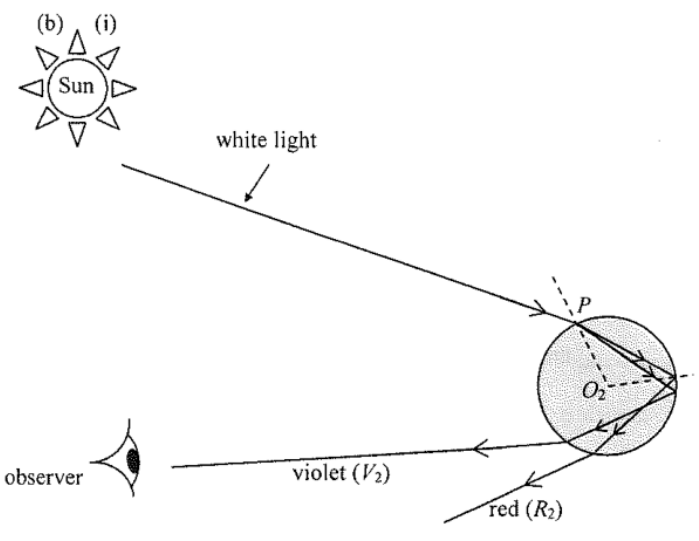
2

4. (a) (i)		2A	
			2
(ii)	Normal reaction (force acting on drum/wall of the drum by the object)	1A	
			1
(b) (i)	$\omega = 78.5 \text{ rad s}^{-1}$ rotation speed = $\frac{78.5}{2\pi} \times 60$ $= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}$	1M 1A	
	Or $T = \frac{2\pi}{\omega} = 0.0800441 \text{ s}$ and $f = \frac{1}{T} = 12.493663 \text{ rev s}^{-1}$ $= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}$	1M 1A	
			2
(ii)	centripetal force $= m\omega^2 r$ $= (2 \times 10^{-5})(78.5)^2(0.25)$ $= 0.03081125 \text{ N} \approx 0.0308 \text{ N}$	1M 1A	
			2
(iii)	$v = r\omega$ $= 0.25 \times 78.5$ $= 19.625 \approx 19.6 \text{ m s}^{-1}$	1M 1A	
			2
(iv)	1	1A	
			1

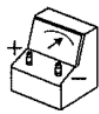
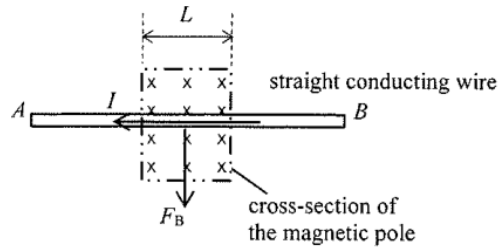
61

		2A	
			2
(ii)	Normal reaction (force acting on drum/wall of the drum by the object)	1A	
			1
(b) (i)	$\omega = 78.5 \text{ rad s}^{-1}$ rotation speed = $\frac{78.5}{2\pi} \times 60$ $= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}$	1M 1A	
	Or $T = \frac{2\pi}{\omega} = 0.0800441 \text{ s}$ and $f = \frac{1}{T} = 12.493663 \text{ rev s}^{-1}$ $= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}$	1M 1A	
			2
(ii)	centripetal force $= m\omega^2 r$ $= (2 \times 10^{-5})(78.5)^2(0.25)$ $= 0.03081125 \text{ N} \approx 0.0308 \text{ N}$	1M 1A	
			2
(iii)	$v = r\omega$ $= 0.25 \times 78.5$ $= 19.625 \approx 19.6 \text{ m s}^{-1}$	1M 1A	
			2
(iv)	1	1A	
			1

5. (a) (i) $a / i / q$	1A 1	
(ii) $c / k / s$	1A 1	
(b) (i) wavelength (4 grids) = $4 \times 50 \text{ cm}$ = 200 cm or 2 m	1A 1	
(ii) speed = frequency $\times$ wavelength frequency = speed $\div$ wavelength = $340 \div 2$ = 170 Hz	1M 1A 2	
(iii) displacement 	2A 2	
(c) Displacement increases, i.e. further to the left.	1A 1	

Solution	Marks	Remarks
6. (a) (i) $\sin r = \frac{\sin 45^\circ}{1.325}$ $r = 32.253454^\circ \approx 32.3^\circ$ (ii) $c = \sin^{-1} \frac{1}{1.325}$ $c = 49.000646^\circ \approx 49.0^\circ$ $r \text{ (or } \angle OBA) = 32.3^\circ < c = 49.0^\circ$, therefore total internal reflection does not occur at point B.	1M 1A 2 1M 1A 1A 3	
(b) (i)  The diagram shows the Sun emitting a beam of white light towards a spherical water droplet. The light enters the droplet at point P. Inside the droplet, the light is dispersed into its constituent colors, with violet light (labeled V₂) and red light (labeled R₂) shown. An observer is shown seeing the violet light. The center of the droplet is labeled O₂. The light path is shown as a solid line, and the normal at point P is shown as a dashed line.	2A 2	
(ii) There is more light energy absorbed by water droplet due to a longer path. Or There is more light energy transmitted to air at the point of reflection as the light undergoes reflection twice.	1A 1A 1A 1A 2	

7. (a)	$R = \rho \frac{l}{A} \Rightarrow 0.50 = \rho \frac{0.20}{8.0 \times 10^{-7}}$ $\rho = 2.0 \times 10^{-6} (\Omega \text{ m})$	1M	
		1A	
(b) (i)	Terminal 3		2
		1A	
(ii)	When switch S is connected to terminal 2, the equivalent resistance across MN $R = \frac{1}{\frac{1}{2.0} + \frac{1}{0.5} + \frac{1}{0.5+0.5+0.5}}$ $= 0.315789 \Omega \approx 0.316 \Omega$		1
		1M	
(iii)	(p.d. across R_2 is) greater than (p.d. across R_1)	1A	
			2
(iv)	zero as no p.d. across CF / no current in the coil.		1
		1A	
			2

Solution		Marks	Remarks								
8. (a)		1A									
		1									
(b)	 straight conducting wire cross-section of the magnetic pole	3A									
		3									
(c)	<table><tr><td></td><td>magnetic force F_B</td></tr><tr><td>using a stronger magnet of the same dimensions</td><td>increase</td></tr><tr><td>using another support such that the position of the horizontal wire is slightly lowered</td><td>unchanged</td></tr><tr><td>using a longer wire and the current is kept unchanged</td><td>unchanged</td></tr></table>		magnetic force F_B	using a stronger magnet of the same dimensions	increase	using another support such that the position of the horizontal wire is slightly lowered	unchanged	using a longer wire and the current is kept unchanged	unchanged	1A	
		magnetic force F_B									
	using a stronger magnet of the same dimensions	increase									
	using another support such that the position of the horizontal wire is slightly lowered	unchanged									
using a longer wire and the current is kept unchanged	unchanged										
		1A									
		1A									
		3									
(d) (i)	(Switch on the power source.) Adjust the power source to deliver a small current, say, 0.5 A to the wire and record the ammeter and the balance readings. Adjust the power source to increase the current stepwise (0.5 A) and record the corresponding balance reading.	1A									
		1A									
		2									
(ii)	The balance reading increases linearly with the current.	1A									
		1									
(iii)	The graph does not pass through the origin because the balance has not been reset to zero initially before the experiment.	1A	Accept: because of the weight of the support / wire								
		1									

9. (a)	(i)	alpha particle / α	1A	$N = N_0 e^{-\lambda t} \quad \text{and} \quad \lambda = \frac{\ln 2}{t_{1/2}}$ $\lambda = \frac{\ln 2}{t_{1/2}} = 4.9159 \times 10^{-11} \text{ yr}^{-1}$ $1 - \frac{N}{N_0} = 1 - e^{-\frac{\ln 2}{t_{1/2}} t}$ $= 1 - e^{-\frac{\ln 2}{1.41 \times 10^{10}} (10)}$ $\approx 4.9 \times 10^{-10}$
			1	
	(ii)	6 α -decays, 4 β -decays	1A	
		$4 n_\alpha = 232 - 208$	1A	
		$2 n_\alpha + (-1) n_\beta = 90 - 82$	2	
		$n_\alpha = 6$		
		$n_\beta = 4$		
	(b)	$\frac{N}{N_0} = \left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}} = \left(\frac{1}{2}\right)^{\frac{10}{1.41 \times 10^{10}}}$ $\text{Portion} = 1 - \frac{N}{N_0} = 1 - \left(\frac{1}{2}\right)^{\frac{10}{1.41 \times 10^{10}}}$ $= 4.9159 \times 10^{-10} \approx 4.9 \times 10^{-10} \text{ (2 sig. fig.)}$	1M	
			1A	
			2	
(c)	(i)	Aluminium body blocks alpha and/or beta particles emitted from thoriated glass.	1A	
		Or		
		Short range of α particles	1A	
			1	
	(ii)	Thorium-232 has an extremely long half-life.	1A	
		Or		
		As part (b) suggests, only an extremely small fraction (4.9×10^{-10}) of thorium decaying in 10 years.	1A	1